

Assignment Code: DS-AG-028

NLP Introduction & Text Processing | Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What is Computational Linguistics and how does it relate to NLP?

Answer:

Computational Linguistics (CL) is a field that combines **linguistics and computer science** to study how human language can be represented, analyzed, and processed using computers. It focuses on developing computational methods to understand the structure and meaning of human languages.

Natural Language Processing (NLP) is a practical application of computational linguistics and artificial intelligence. NLP uses computational techniques to enable computers to **understand, interpret, generate, and communicate using human language**.

Relationship between CL and NLP:

Computational Linguistics provides the linguistic theories and knowledge about **grammar, syntax, semantics, and language structure**, while NLP applies these ideas to build real-world systems such as **chatbots, machine translation, speech recognition, sentiment analysis, and text summarization**.

Question 2: Briefly describe the historical evolution of Natural Language Processing.

Answer:

The evolution of **Natural Language Processing (NLP)** can be divided into several important stages:

1. **1950s–1960s: Early NLP and Rule-Based Systems**

NLP began with attempts to make computers understand human language. Early systems relied mainly on **hand-written rules, dictionaries, and grammar**. The famous **ELIZA** program (1960s) simulated conversation using simple pattern matching.

2. **1970s–1980s: Knowledge-Based Approach**

Researchers developed systems that used large collections of linguistic rules and knowledge bases. These systems could perform tasks such as **language understanding, question answering, and machine translation**, but they were difficult to maintain.

3. **1990s: Statistical NLP**

NLP shifted toward **statistical and machine-learning methods**. Instead of relying only on manually written rules, systems learned patterns from large text datasets. Techniques such as **Hidden Markov Models (HMMs)** and probabilistic language models became popular.

4. **2000s: Machine Learning and Large-Scale Data**

With increasing computing power and availability of large datasets, machine-learning algorithms such as **Support Vector Machines, Naïve Bayes, and Conditional Random Fields** were widely used for NLP tasks.

5. **2010s: Deep Learning Era**

Neural networks significantly improved NLP performance. **Word embeddings, RNNs, LSTMs, and CNNs** enabled computers to capture more complex relationships between words and sentences.

6. **Late 2010s–2020s: Transformers and Large Language Models**

The introduction of the **Transformer architecture** led to major advances in NLP. Models such as **BERT, GPT, and T5** achieved strong performance in language understanding and generation. Today, **Large Language Models (LLMs)** can perform tasks such as answering questions, summarizing text, translating languages, generating content, and engaging in conversations.



Question 3: List and explain three major use cases of NLP in today's tech industry.

Answer:

Natural Language Processing (NLP) is widely used in the technology industry to help computers understand and process human language. Three major use cases are:

1. **Chatbots and Virtual Assistants**

NLP enables chatbots and virtual assistants to understand user questions and provide relevant responses. Examples include customer-support chatbots, voice assistants, and AI conversational systems.

2. **Machine Translation**

NLP is used to automatically translate text from one language to another. Applications such as Google Translate use NLP techniques to understand the meaning and generate translations between different languages.

3. **Sentiment Analysis**

NLP can analyze text to identify the **emotion or opinion** expressed by a person, such as positive, negative, or neutral sentiment. Companies use sentiment analysis to analyze customer reviews, social-media posts, and feedback.

Question 4: What is text normalization and why is it essential in text processing tasks?

Answer:

Text normalization is the process of converting text into a **standard and consistent format** before further processing. It reduces unnecessary variations in text so that NLP systems can process it more effectively.

Common text normalization techniques include:

- **Converting text to lowercase:** "Hello" → "hello"
- **Removing unnecessary punctuation:** "Hello!" → "Hello"
- **Removing extra spaces**
- **Expanding abbreviations:** "can't" → "cannot"
- **Handling special characters and numbers**
- **Stemming or lemmatization** when required

Why is Text Normalization Essential?

1. **Improves consistency:** Different forms of the same word are converted into a common format.
2. **Reduces complexity:** It decreases the number of unique words or tokens that a model needs to process.
3. **Improves accuracy:** Clean and consistent text helps NLP models perform better.
4. **Makes text easier to analyze:** Tasks such as sentiment analysis, classification, and information extraction become more effective.

Example:

Without normalization: "HELLO", "Hello!", "hello"

After normalization: "hello", "hello", "hello"

Question 5: Compare and contrast stemming and lemmatization with suitable examples.

Answer:

Stemming and **lemmatization** are NLP techniques used to reduce words to their basic or root form. However, they work differently.

Feature	Stemming	Lemmatization
Meaning	Removes prefixes/suffixes to obtain a root form	Converts a word to its meaningful dictionary base form
Method	Uses simple rules or algorithms	Uses vocabulary and linguistic analysis
Speed	Faster	Slower
Accuracy	Less accurate	More accurate
Output	May produce an invalid word	Usually produces a valid word
Example	<i>studies</i> → <i>studi</i>	<i>studies</i> → <i>study</i>
Example	<i>running</i> → <i>run</i>	<i>running</i> → <i>run</i>
Example	<i>better</i> → <i>better</i>	<i>better</i> → <i>good</i>

Example

Consider the words:

"The students are studying their studies."

- **Stemming:** *students* → *student*, *studying* → *studi*, *studies* → *studi*
- **Lemmatization:** *students* → *student*, *studying* → *study*, *studies* → *study*

Question 6: Write a Python program that uses regular expressions (regex) to extract all email addresses from the following block of text:

“Hello team, please contact us at support@xyz.com for technical issues, or reach out to our HR at hr@xyz.com. You can also connect with John at john.doe@xyz.org and jenny via jenny.clarke126@mail.co.us. For partnership inquiries, email partners@xyz.biz.”

(Include your Python code and output in the code box below.)

Answer:

[Code Link](#)

Question 7: Given the sample paragraph below, perform string tokenization and frequency distribution using Python and NLTK:

“Natural Language Processing (NLP) is a fascinating field that combines linguistics, computer science, and artificial intelligence. It enables machines to understand, interpret, and generate human language. Applications of NLP include chatbots, sentiment analysis, and machine translation. As technology advances, the role of NLP in modern solutions is becoming increasingly critical.”

(Include your Python code and output in the code box below.)

Answer:

[Code Link](#)

Question 8: Create a custom annotator using spaCy or NLTK that identifies and labels proper nouns in a given text.

(Include your Python code and output in the code box below.)

Answer:

[Code Link](#)

Question 9: Using Genism, demonstrate how to train a simple Word2Vec model on the following dataset consisting of example sentences:

dataset = [

"Natural language processing enables computers to understand human language",

"Word embeddings are a type of word representation that allows words with similar meaning to have similar representation",

"Word2Vec is a popular word embedding technique used in many NLP applications",

"Text preprocessing is a critical step before training word embeddings", "Tokenization and normalization help clean raw text for modeling"

]

Write code that tokenizes the dataset, preprocesses it, and trains a Word2Vec model using Gensim.

(Include your Python code and output in the code box below.)

Answer:

[Code Link](#)

Question 10: Imagine you are a data scientist at a fintech startup. You've been tasked with analyzing customer feedback. Outline the steps you would take to clean, process, and extract useful insights using NLP techniques from thousands of customer reviews.

(Include your Python code and output in the code box below.)

Answer:

[Code Link](#)